Mid-term Revenue Risk Assessment Model for Highway Public-Private Partnership Projects Based on Copula-NPVaR

Yujie Huang
Graduate Research Assistant
Shanghai Key Laboratory for Digital Maintenance of Buildings and Infrastructure,
School of Naval Architecture, Ocean & Civil Engineering
Shanghai Jiao Tong University, Shanghai, 200240, China
Email: vegetablemonkey@sjtu.edu.cn

Zhipeng Zhang, Ph.D. (Corresponding Author)
Associate Professor
State Key Laboratory of Ocean Engineering
School of Naval Architecture, Ocean and Civil Engineering
Shanghai Jiao Tong University, Shanghai, 200240, China
Email: zp.zhang@sjtu.edu.cn

Lei Dai, Ph.D.
Associate Professor
State Key Laboratory of Ocean Engineering
School of Naval Architecture, Ocean and Civil Engineering
Shanghai Jiao Tong University, Shanghai, 200240, China
Email: dailei1989@sjtu.edu.cn

Hao Hu, Ph.D.
Professor
State Key Laboratory of Ocean Engineering
School of Naval Architecture, Ocean and Civil Engineering
Shanghai Jiao Tong University, Shanghai, 200240, China
Email: hhu@sjtu.edu.cn

Jie Xue, Ph.D.
Assistant Research Professor
State Key Laboratory of Ocean Engineering
School of Naval Architecture, Ocean and Civil Engineering
Shanghai Jiao Tong University, Shanghai, 200240, China
Email: j.xue@sjtu.edu.cn

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Abstract

Public-Private Partnership (PPP) proves to be an effective approach to solve the financing problems of highway projects. However, there is a lack of study on how to control its high revenue risk on financial returns caused by great uncertainties regarding revenue and cost during a long concession period for both public and private sectors. Although the concept of life-cycle risk management has been emphasized a lot by researchers and practitioners, few studies focused on the mid-term risk assessment model and there is still no well-developed mid-term risk assessment mechanism. To fill this research gap, this paper attempts to establish a mid-term revenue risk assessment model combining Copula and NPV-at-Risk method. Instead of viewing the annual revenue and cost of the project as independent variables, this study applies Copula functions to build joint distribution of them in the simulation, which is the main improvement made on existing NPV-at-Risk models. The Copula-NPVaR model is applied in a numerical case study based on a highway project in China. The results indicate that by means of Copula, the simulated NPVs, under the more reasonable stochastic assumptions of random variables, will approach to the real value. This model can be a useful tool to assess the profitability and financial sustainability of the project in operation period providing the stakeholders with more reliable and quantitative reference on the revenue risk. This study offers both theoretical foundations and practical suggestions to practitioners that will ultimately promote the adaptive and agile management of highway PPP projects.

Keywords: Highway PPP Projects; Revenue Risk; Mid-Term Risk Assessment; Infrastructure Finance

1. Introduction

Financing issue of transportation infrastructures has been a controversial topic in developing countries. With the rapid development of highway network since 1980s in China, fiscal investment from the government cannot fulfill the huge fund demand for highway construction. Furthermore, debt financing proves to be an unsustainable way since the local-government debt has climbed to roughly 21.5% of China's nominal GDP as of 2019 (*1*). Consequently, PPP (Public-Private Partnership) mode is greatly encouraged in highway and other infrastructures by Chinese central government in order to solve the financing problems and increase the efficiency and quality of public services. By absorbing the fund and expertise of social sectors, PPP is regarded as a satisfactory solution to the undersupply of infrastructures. Driven by positive policies, China is currently the biggest PPP market in the world, with a total investment of over 2 trillion US dollars (*2*).

PPP is a long-term cooperative agreement between public and private sectors, in which the private sector is responsible for financing and providing public products or services for reasonable revenues (*3*). As the operator of projects, the private sector shares significant part of the revenue risk failing to obtain reasonable revenue during operation. Highway PPP projects, characterized as huge initial investment, long concession period and various uncertainties in traffic demand and cost, typically involve high revenue risk (*4*). Therefore, many research endeavors have focused on the risk assessment for road PPP projects to evaluate the potential revenue risk and aid decision-making.

Due to the long concession period, instead of only ex-ante analysis, a life-cycle risk management framework for PPP infrastructure projects should be developed to realize value-for-money and balance of interest between stakeholders (*5*). According to (*6*), over 80% of the highway PPP projects in China is launched after 2014, which means most of them are at their preliminary stage and confronted with unpredicted revenue risk due to uncertainties in revenue and cost forecast. In order to objectively reflect the quality of service, efficiency and emerging risks and to make reasonable adjustment on PPP agreement, the project managers are required to conduct periodic assessments (*7*). Thus, mid-term revenue risk assessments will help the project managers to understand current financial status of the project. However, one of the critical questions to be answered is how to model the potential revenue risk in mid-term assessment. The absence of such models will undermine the recognition of emerging revenue risk and hinder the project managers to make right decisions on risk mitigation or allocation mechanism.

The primary objective of this paper is to propose a mid-term revenue risk assessment model, and then apply it based on a Chinese highway PPP project case. The results may provide a guideline and foundation for the establishment of a standard mid-term revenue risk assessment framework, which paves a step in adaptive and agile project management.

2. Literature Review

2.1 Life-cycle Risk Management for PPP Projects

Numerous studies emphasized adopting life-cycle approaches in risk management for PPP. Le et al. (*8*) and Liu et al. (*9*) highlighted the unique, dynamic, and complex nature of risk factors in PPP projects, attributable to their substantial initial investments and long concession periods. Conversely, the absence of life-cycle perspective in PPP management will result in producing suboptimal service quality and more unexpected risks (*10*).

Figure 1 illustrates a life-cycle framework proposed by Zou et al. (*5*) aiming to balance the interest of stakeholders involved. This dynamic process encompasses risk identification, analysis, allocation, and monitoring, from feasibility analysis continuing through the operation and transfer phases. It is apparent that risk assessments have strong relationship with other critical steps. For instance, typical dynamic risk allocation mechanisms in transportation PPP projects include minimum revenue guarantees and maximum revenue caps (*11-12*). To mitigate potential losses within the risk tolerance of stakeholders, risk assessment is supposed to be conducted. Based on the assessment results, tools like game theory can be applied to analyze the behavioral dynamics to achieve the optimal risk allocation (*13*).

Risk assessment from a life-cycle perspective involves not only ex-ante assessment but also mid-term and dynamic assessment (*14*). Mid-term assessment is particularly crucial in periods involving negotiation, renegotiation, and risk allocation adjustments to achieve optimal value for money. Due to long concession periods of transportation PPP projects, multiple mid-term assessments should be conducted as response to the unexpected risks. According to the policy issued by the Ministry of Finance in China, the monitoring institutions of PPP project are supposed to carry out mid-term assessments every 3-5 years, analyzing the operation situation, verifying the realization of value for money and assessing the unexpected risks. However, there’s a lack of well-developed mid-term risk assessment model that clarifies the content, method and requirement (*15*).

As for highway PPP projects, the financial distress is mainly driven by inaccurate forecast on traffic volume and operating cost, defined as revenue risk (*16*). Therefore, an efficient mid-term revenue risk assessment model is needed to evaluate and monitor the potential loss of highway PPP projects caused by revenue risk.

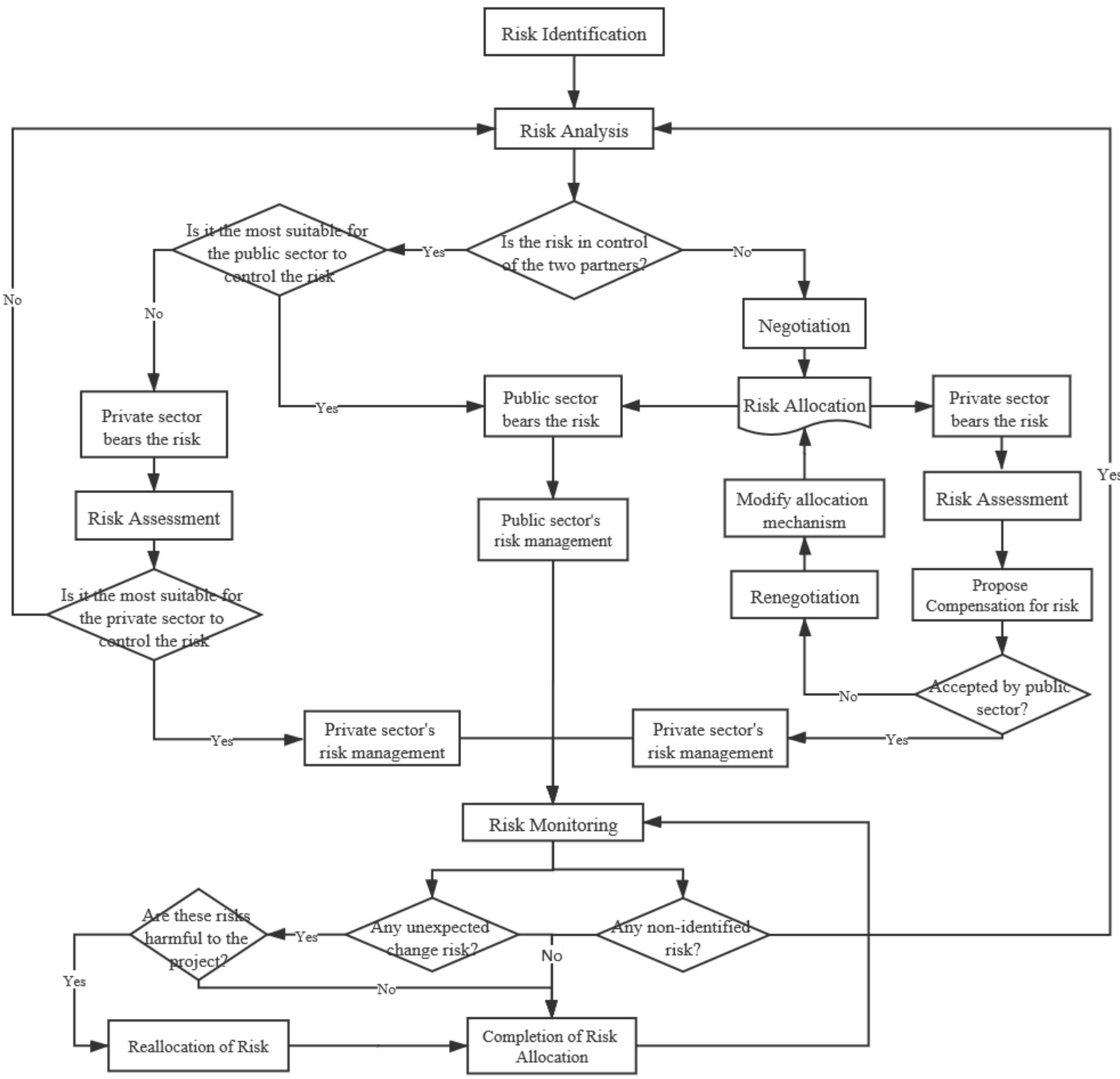

Figure 1. A Life-cycle Risk Management Framework for PPP Projects (Adapted from (*5*))

2.2 Revenue Risk Assessment Models

The revenue risk of highway PPP projects encompasses the probability of actual revenue falling below forecasted due to insufficient traffic flow, operation cost overrun, debt and the adequate rate of return (*17*). Generally, the Discounted Cash Flow (DCF) Model and the Non-Discounted Cash Flow Model are commonly employed to evaluate project revenue risk. The former one stresses the time value of money and calculates the net present value (NPV) using a discount rate, while the latter focuses on the payback period or rate of return (*4*). Given that the concession period of highway PPP project is typically longer than twenty years, DCF models proves to be a suitable choice for risk assessment (*18*). However, since traditional DCF models failed to adequately reflect the market volatility, researchers investigated stochastic models to incorporate the uncertainty (*16*). Two main stochastic tools are applied in revenue risk assessment for highway PPP projects: NPV-at-Risk and real option.

NPV-at-Risk (NPVaR) was proposed by Ye and Tiong (*19*) to portray the risk profile via NPV within a confidence interval, which was an application of financial Value-at-Risk (VaR) into project management domain (*20*). Based on the previous prototype, Zhu et al. (*21*) established a stochastic model for BOT scheme, accounting for the impact of risk attitude of stakeholders. Kumar et al. (*4*) proposed a standard financial risk analysis model based on NPVaR to assess 30 real highway PPP projects in India. However, these models failed to utilize the operating data available for mid-term assessment. Namely, the stochastic assumptions within these models were mainly based on empirical hypothesis from previous project cases, which may not accurately reflect the real-time situation of the project being appraised.

Real option theory is another feasible tool especially when risk sharing mechanisms like minimum revenue guarantee (MRG) and maximum revenue cap (MRC) are applied to the project. These forms of government support or obligations can be interpreted as real option since they are triggered when some pre-specified conditions are met (*22-24*). In real option revenue risk assessment models, the traffic revenue was assumed to follow a Geometric Brownian motion, which also failed to tell the trend closest to the real situation.

Another problem exists in the multivariate stochastic assumptions of above studies: the variables were set to be independent, or covariance values between them are empirically set without solid evidences. Actually, the relevance between variables does exist and can be derived in statistical method if enough operation data is available in concession period. Specifically, the maintenance and operation cost will probably increase when increasing traffic flow brings more revenue. The accuracy of simulation results will be undermined if this relationship is neglected in revenue risk assessment.

2.3 Copula-based Method

Copula is a function used to describe the dependency between random variables. Dong et al. (*25*) built a Copula-based model for infrastructure project lenders involved in a portfolio of assets to measure the credit risk. Compared with previous studies on one single project investment, Copula helps to assess the joint default risk in project financing. Chakkalakal et al. (*26*) assessed the risk-return characteristics of transport infrastructure portfolio from the perspective of institutional investors based on t-Copula-CVaR (Conditional Value at Risk) model and the results supported the use of this novel risk assessment tool incorporating non-normal distributions to represent the multivariate dependence structures. The major advantage of Copula-based model is that it circumvents the restrictive stochastic assumptions of NPVaR models and real option models (*27*), and it generates joint probability distribution in multivariate models.

This paper proposes a Copula-NPVaR model for mid-term revenue risk assessment of highway PPP projects, which answers two questions: how to make use of cash flow information derived in previous period in mid-term assessment and how to build a reasonable joint stochastic assumption of multivariate structure in the simulation. To clarify the research gap, a summary of revenue risk assessment models for transport infrastructure project is presented in Table 1.

Table 1 A Summary of Revenue Risk Assessment Models

Model Category & Stochastic Assumption	Representative Author	Remarks	Area of Application
NPV-at-Risk *Stochastic Assumption*: Multiple random variables, such as traffic volume, operation cost and toll rate, are set to be independent, or set a covariance between jointly distributed random variables.	Ye and Tiong (*19*)	Develop NPV-at-Risk method and provide a better decision for privately financed infrastructure projects investment.	Ex-ante revenue risk or investment risk analysis
	Zhu et al. (*21*)	Propose a stochastic BOT model which considers the impact of risk and risk attitudes of investors to get a more reasonable concession interval.	
	Kumar et al. (*4*)	Apply NPV-at-Risk model to 30 practical BOT highway projects to identify the critical risk and discuss the risk mitigation mechanism.	
Real Option *Stochastic Assumption:* Traffic flow is thought to be the main factor contributing to the volatility of assessment model. The variable of	Cheah and Liu (*22*)	Treat the governmental support as real options and evaluate the value of options through Monte Carlo simulation of DCF model.	Evaluation of the revenue risk under risk allocation mechanism like minimum revenue

traffic volume or traffic revenue is set to follow the Geometric Brownian motion.	Iyer and Sagheer (*23*)	Propose a ‘traffic band’ combining traffic floor and traffic cap and evaluate this real option using binomial lattice method.	guarantee and maximum revenue cap
	Buyukyoran and Gundes (*24*)	Identify an optimum upper and lower boundary of MRG and MRC based on real option method.	
Copula-NPVaR (applied in this paper) *Stochastic Assumption:* A Copula function is built according to the real cash flow information obtained in operation period to present the joint distribution of income and cost.		Establish a mid-term assessment model making use of cash flow information gained in operation period and apply Copula to reveal the relationship between variables.	Mid-term revenue risk assessment model

3. Model Establishment

3.1 Model Framework

This subsection introduces the framework of mid-term revenue risk assessment model proposed in this paper, clarifying the objective, methodology, input and output of the model.

The model aims to evaluate the revenue risk level in a mid-term assessment by utilizing the cash flow information of cash inflows and outflows generated from the project appraised. Since the evaluation mainly focuses on predicting the NPV under the influence of the uncertainty and fluctuation of cash flow and without risk allocation mechanisms like MRG and MRC, NPV-at-Risk could be an intuitive method to realize the goal. Different from the experience-based stochastic assumptions in previous studies, for mid-term assessment, the probability distribution assumptions of random variables can be derived from the practical data during the operation period. Copula is an important tool in this model to produce joint distribution of relevant variables and make the stochastic assumption more reasonable.

Consequently, the assessment model is designed to be a Discounted Cash Flow model with relevant random variables, and Monte Carlo simulation can be applied to work out the distribution of NPV and NPV-at-Risk (NPV at different confidence levels).

The above discussed model framework is illustrated in Figure. 2.

3.2 Discounted Cash Flow Model

In the case study of Chinese highway PPP projects, Zou et al. (*5*) concluded that an unfavorable feature of toll road compared to other infrastructures is the great uncertainty regarding the cost and income, on which the model in this paper focuses. Discounted cash flow (DCF) model is applied to obtain the net present value (NPV) of operation period as shown in Equation (1). In China, a special purpose vehicle (SPV) of highway PPP project gets over 70% of the financing from bank lender and the rest is equity provided by stakeholders. To take into account both the rate of return of stakeholders and the debt interest, the weighted average cost of capital (WACC) can be used as discount rate for the project as Equation (2) (*28*).

$$NPV_O = \sum_{t=1}^{T} \frac{R_t - C_t}{(1 + WACC)^t} \tag{1}$$

$$WACC = P_E \times R_E + P_D \times R_D \times (1 - T_C) \tag{2}$$

where NPV_O is the net present value generated from all the operating activities; the length of concession period is T; the annual income in year t is assumed to be R_t, while the annual operating cost is C_t; P_E is the percentage of financing that is equity and P_D is that of debt; R_E refers to the cost of equity, in other words, the expected rate of return of private sectors; R_D is the cost of debt, the interest rate; T_C is the corporate tax rate.

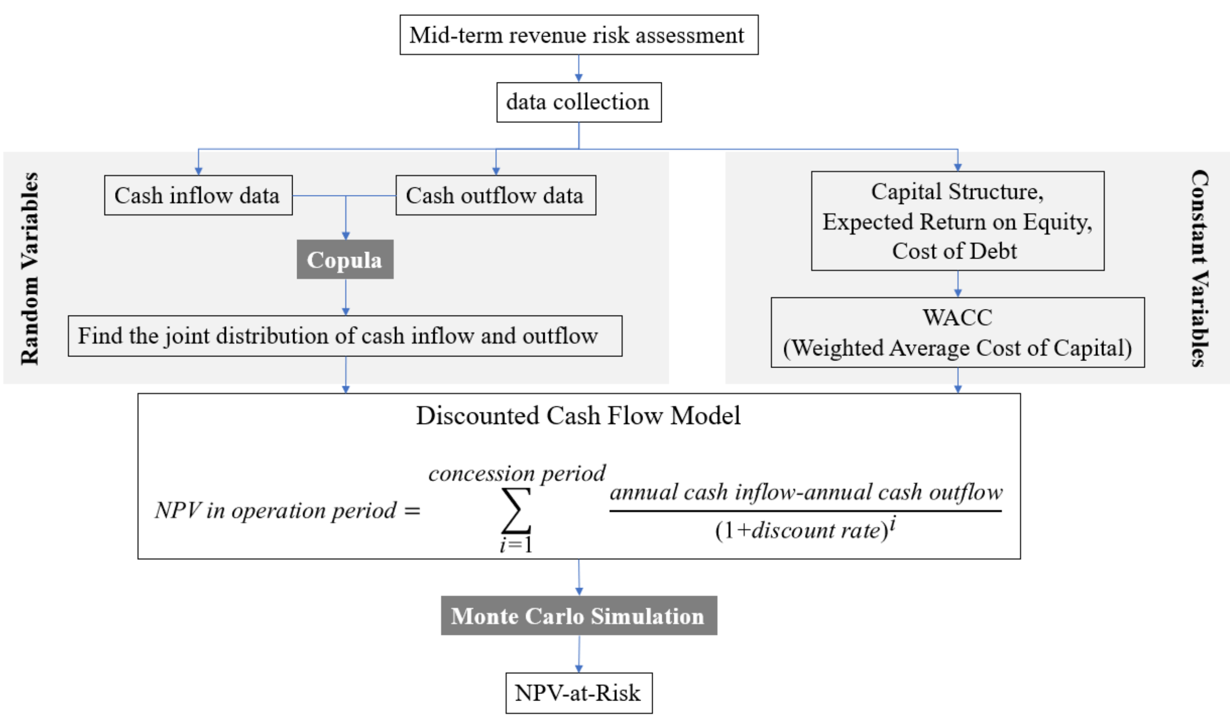

Figure 2. Mid-term Revenue Risk Assessment Model Framework

The costs in construction period are not taken into account because they are sunk costs from the perspective of mid-term assessment in operation period. The annual income of highway projects consists of toll income and billboard rents income. However, according to the annual reports of Chinese highway companies, the billboard rents only account for 1% of the annual income (*29*). Thus, the annual income R_t can be regarded as a variable entirely dependent on the traffic flow. As for the annual cost of highway, the main divisions include labor cost, facility expenditure, maintenance expenditure and interest payment. Since the interest payment is actually incorporated into the WACC, there is no need to consider it again. The remaining part of cost, in Mamdoohi (*30*) 's research, has been found to vary closely with the traffic, which is consistent with the cognition that in any business the cost of production will be influenced by the variations of the volume. Therefore, the sequences of R_t and C_t should be interdependent random sequences here to represent both their volatility and relationship.

3.3 NPV-at-Risk

NPV-at-risk (*NPVaR*) is a valuation of project investment, which calculates the probable minimal NPV given a confidence level (*19*). In this paper, $NPVaR_{1-\alpha}$ (NPV-at-Risk at the confidence level of $1-\alpha$) is mathematically defined as the infimum of the set of NPV whose cumulative distribution function value is larger than α.

$$NPVaR_{1-\alpha} = \inf\{x \epsilon \mathbb{R}, F_{NPV}(x) > \alpha\} \tag{3}$$

F_{NPV} is the cumulative distribution function of NPVs as shown in Figure 3. Then, the key task is to build the distribution function of NPV based on the cash flow data in operation period.

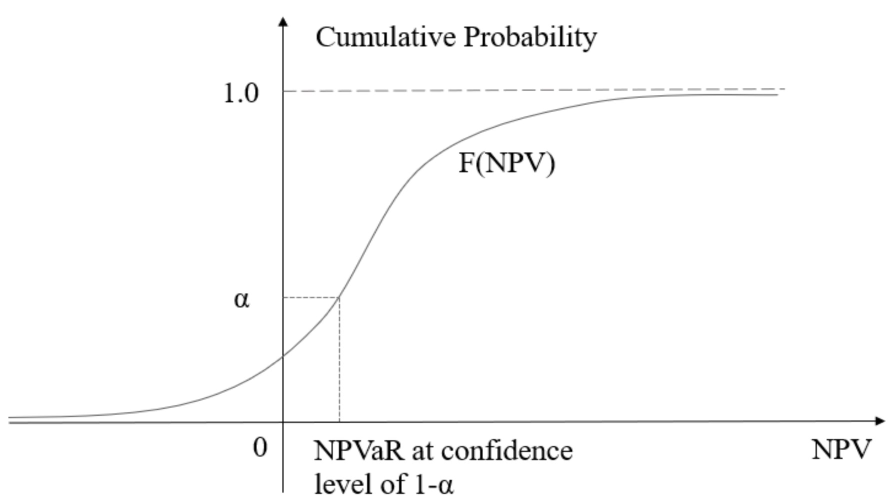

Figure 3. Calculate the NPVaR Based on Cumulative Distribution Function of NPVs.

3.4 Copula

In order to get the NPV distribution, stochastic assumptions on the random variables in Equation (1) should be made. The relationship between annual income and cost has been discussed above, so a joint distribution of them should be established. This section introduces Copula theory, which is applied to obtain the joint distribution of annual income and cost.

According to Sklar's Theorem, if $H_{X,Y}(x,y)$ is a joint distribution function with marginal distributions $F_X(x)$ and $G_Y(y)$, then there exists a Copula function $C(u,v)$ such that for all u,v in $\mathbf{I}^2$ ($[0,1]^2$),

$$H_{X,Y}(x,y) = C\{F_X(x), G_Y(y)\} \tag{4}$$

Given that $F_X(x)$ and $G_Y(y)$ are continuous functions, the uniqueness of $C(u,v)$ can be established. However, in practical scenarios, determining this unique Copula function becomes highly challenging. Fortunately, researchers have introduced various Copula functions that can effectively replace $C(u,v)$, offering more practical and manageable alternatives.

Among families of Copulas, Archimedean copulas are more popular and widely used in applications because of their simple form and nice property (*31*). Archimedean copulas are copulas that can be expressed in terms of:

$$C(u,v) = \varphi^{[-1]}(\varphi(u) + \varphi(v)) \tag{5}$$

where φ is a strictly decreasing, continuous function and is known as the generator of an Archimedean copula. $\varphi^{[-1]}$ is the 'pseudo inverse' defined as:

$$\varphi^{[-1]}(t) = \begin{cases} \varphi^{-1}(t), & 0 \le t \le \varphi(0) \\ 0, & \varphi(0) \le t \le \infty \end{cases} \tag{6}$$

The generators φ of Archimedean family usually have only one parameter θ in the function and we can obtain the value of θ through Kendall rank correlation coefficient, also known as Kendall's tau (τ). Kendall's τ is a measure of the rank correlation between two quantities. If a set of observations of joint random variables X and Y are obtained, the Kendall's τ of this sample is defined as:

$$\tau_s = P\left(\left(x_i - x_j\right)\left(y_i - y_j\right) > 0\right) - P\left(\left(x_i - x_j\right)\left(y_i - y_j\right) < 0\right) \tag{7}$$

Given pairs $(x_i,\ y_i)$ and $\left(x_j,\ y_j\right)$ in observations, if $\left(x_i - x_j\right)\left(y_i - y_j\right) > 0$, they are defined as concordant pairs, and if $\left(x_i - x_j\right)\left(y_i - y_j\right) < 0$, then they are discordant pairs. There are totally $\binom{n}{2} = \frac{n(n-1)}{2}$ ways of picking 2 pairs out of n. Therefore, τ_s can be obtained from Equation (8).

$$\tau_s = \frac{2(c-d)}{n(n-1)} \tag{8}$$

where c means the number of concordant pairs and d is the number of discordant pairs.

Above is the measure of observations, and, also, for an Archimedean Copula function with a generator φ_θ, its Kendall rank correlation coefficient τ_θ can be obtained from:

$$\tau_\theta = 1 + 4\int_0^1 \frac{\varphi_\theta(u)}{\varphi'_\theta(u)} du \tag{9}$$

Given a generator of Archimedean family, let $\tau_\theta = \tau_s$, and then the value of parameter θ can be derived. Thus, the Copula function is determined and we can generate random values of operating revenue and cost. Table 2 lists four commonly used Archimedean Copulas and in the following section of this paper, they will be applied to generate the joint random variables.

Table 2 Four Commonly Used Archimedean Copulas

Copula	Generator φ_θ	$\theta \in$	$C_\theta(u,v)$	τ_θ
Clayton	$\frac{1}{\theta}(t^{-\theta}-1)$	$[-1,\infty)\backslash\{0\}$	$[\max(u^{-\theta}+v^{-\theta}-1,0)]^{-\frac{1}{\theta}}$	$\frac{\theta}{\theta+2}$
Frank	$-\ln(\frac{e^{-\theta t}-}{e^{-\theta}-1}$	$(-\infty,\infty)\backslash\{0\}$	$-\frac{1}{\theta}\ln[+\frac{(e^{-\theta u}-1)(e^{-\theta v}-1}{e^{-\theta}-1}$	$1+\frac{4}{\theta}\left(\frac{1}{\theta}\int_0^\theta \frac{t}{e^t-1}dt-1\right)=0$
Ali-Mikhail-Haq	$ln\frac{1-\theta(1-}{t}$	$[-1,1)$	$\frac{uv}{1-\theta(1-u)(1-v)}$	$\left(\frac{3\theta-2}{\theta}\right)-\frac{2}{3}\left(1-\frac{1}{\theta}\right)^2\ln(1-\theta)$
Gumble Hougard	$(-\ln t)^\theta$	$[1,\infty)$	$\exp(-[(-\ln u)^\theta+(-\ln v)^\theta]^{1/\theta})$	$\frac{\theta-1}{\theta}$

4 Numerical Calculation with Simulation

4.1 Random Variables Generation

Lu et al. (*32*) investigated the growth pattern of truck traffic volume based on data of California highways and found that both compound growth and linear growth functions can model the growth trends of highway's traffic volume. Since the volatility of toll rates in China is strictly controlled by the government, the toll rate can be regarded as constant variable, and thus we can treat the annual income as linear growth or compound growth. Also, we assume that the annual cost has the same growth pattern as annual income. At first, to simplify the calculation, we assume the actual growth is consistent with linear growth pattern and verify this assumption through following steps. Let R_t^r and C_t^r be the actual annual income and cost in the year t in operation period. Then, obtain the first-order difference as $\Delta R_t^r = R_{t+1}^r - R_t^r$ and $\Delta C_t^r = C_{t+1}^r - C_t^r$. ΔR_t^r and ΔC_t^r should be stationary sequence due to the linear growth assumption of R_t^r and C_t^r, which means their values do not depend on the time. In the real case simulation, ADF (Augmented Dickey–Fuller) Test can be applied to verify the stationarity of this sample. If the sample passes the stationary test, then, $(\Delta R, \Delta C)$ can be the bivariate random vector to be generated in following step. If it does not pass the test, then consider the annual growth rate as $\Delta R_t^r = \frac{R_{t+1}^r}{R_t^r} - 1$ and $\Delta C_t^r = \frac{C_{t+1}^r}{C_t^r} - 1$. The following steps are based on the situation that the first-order differences of income and cost are stationary.

Bivariate Archimedean copula can be employed to generated relevant random variables (*33*). This simulation procedure was based on following theorem:

(U_1, U_2) is a bivariate random vector with uniform marginals. The joint distribution function of u_1, u_2 is of the Archimedean copula form $C(u_1, u_2) = \varphi^{[-1]}(\varphi(u_1) + \varphi(u_2))$ with a particular generator φ.Then we define random variables $S = \frac{\varphi(U_1)}{\varphi(U_1)+\varphi(U_2)}$ and $T = C(U_1, U_2)$. The joint distribution function of S and T can be characterized by:

$$H(s,t) = P(S \leq s, T \leq t) = s \times K_C(t) \tag{10}$$

where $K_C(t) = t - \frac{\varphi(t)}{\varphi'(t)}$. S and T are independent variables and S is uniformly distributed on $(0,1)$.

Suppose $(\Delta R, \Delta C)$ has a bivariate distribution function based on a two-dimensional Archimedean copula with generator φ and their marginal distributions are F_R and F_C which can be obtained according to the actual data in operation period. The simulation procedure is as follow:

(1) Generate independent random variables s and w, and both of them follow uniform distribution $U(0,1)$.

(2) Let $t = {K_C}^{-1}(w)$.

(3) Let $u = \varphi^{-1}[s\varphi(t)]$ and $v = \varphi^{-1}[(1-s)\varphi(t)]$.

(4) And the desired simulated value $(\Delta R, \Delta C) = ({F_R}^{-1}(u), {F_C}^{-1}(v))$.

4.2 Choice of a Copula

Suppose it is year n in operation period and the first-order difference pairs $(\Delta R_1^r, \Delta C_1^r)$, $(\Delta R_2^r, \Delta C_2^r)$, …, $(\Delta R_{n-1}^r, \Delta C_{n-1}^r)$ of n years are available. Let Kendall's tau of sample equals to that of one Copula, $\tau_\theta = \tau_s$, and then the value of θ, and the generator φ_θ is determined.

Given this φ_θ, the simulated pairs $(\Delta R_1^s, \Delta C_1^s)$, $(\Delta R_2^s, \Delta C_2^s)$, …, $(\Delta R_{n-1}^s, \Delta C_{n-1}^s)$ of

n years in operation period can be obtained through the above procedure in Subsection: Random Variables Generation. To measure the accuracy of simulation, we compare the net present value based on simulated data $NPV_{[1,n]}^{s}$ and that of practical data $NPV_{[1,n]}^{r}$ for these n years.

$$NPV_{[1,n]}^{s} = \sum_{t=1}^{n} \frac{R_t^s - C_t^s}{(1+d)^t} \tag{11}$$

$$R_t^s = R_1^r + \sum_{i=1}^{t-1} \Delta R_i^s \tag{12}$$

$$C_t^s = C_1^r + \sum_{i=1}^{t-1} \Delta C_i^s \tag{13}$$

$$NPV_{[1,n]}^{r} = \sum_{t=1}^{n} \frac{R_t^r - C_t^r}{(1+d)^t} \tag{14}$$

Repeat the simulation process for N times, and then $NPV_{[1,n],1}^{s}, NPV_{[1,n],2}^{s}, \dots, NPV_{[1,n],N}^{s}$ are obtained. To verify the effect of this simulation, we define two indexes, the mean value and the degree of deviation noted as M and D:

$$M = \frac{1}{N} \sum_{i=1}^{N} NPV_{[1,n],i}^{s} \tag{15}$$

$$D = \sqrt[N]{\prod_{i=1}^{N} \left| \frac{NPV_{[1,n],i}^{s} - NPV_{[1,n]}^{r}}{NPV_{[1,n]}^{r}} \right|} \tag{16}$$

M measures the average level of simulated numbers and D measures the dispersion of them. The simulation is more accurate when M is close to $NPV_{[1,n]}^{r}$ and D is small enough. Essentially, values of M and D are determined by generator φ_θ. Consequently, an appropriate Copula can be chosen according to the value of M and D.

4.3 Calculation of NPV-at-Risk

Once the Copula is determined, the first-order difference pairs of the rest of operation period, $(T-n)$ years, can be generated and noted as $(\Delta R_{n+1}^s, \Delta C_{n+1}^s)$, $(\Delta R_{n+2}^s, \Delta C_{n+2}^s), \dots,$ $(\Delta R_T^s, \Delta C_T^s)$. Then a simulated NPV of mid-term assessment in year n, noted as $NPV_{mid,n}^{S}$, can be worked out as:

$$NPV_{mid,n}^{s} = NPV_{[1,n]}^{r} + NPV_{[n+1,T]}^{s} \tag{17}$$

$$NPV^s_{[n+1,T]} = \sum_{t=n+1}^{T} \frac{R^s_t - C^s_t}{(1+d)^t} \tag{18}$$

where NPV^r_n, R^s_t, C^s_t is shown in Equation (12), (13) and (14).

With the help of computer programing, repeat this process and get a large number of $NPV^S_{mid,n}$. Finally, plot the distribution of them and find the $NPVaR_\alpha$ according to figure 3.

5 Application of the Copula-NPVaR: A Numerical Example

5.1 Data Processing

In this section, a case of Chinese highway project is introduced and applied to verify the above assessment method. All the equation solving and calculation processes are done with MATLAB R2018a.

Chang Jiu highway in Jiang Xi Province links the capital city Nanchang and a prefecture-level city Jiujiang. Since the integrated regional urban development of these two cities has been greatly promoted by local government, another traffic link, the intercity railway, has also been built between these two cities, which shares part of the traffic volume and influences the operation revenue of Chang Jiu highway. Therefore, it is necessary to carry out mid-term risk assessment during the operation period for this highway project. The equity accounts for 30% of the financing and the other 70% is debt. The debt interest rate is 5% and the rate of return for the investors is set to be 8%. Thus, the WACC is 5% with a corporate tax rate of 25%.

The first step is to select the time point (year n) for mid-term assessment. n should not be too small, or the appraiser cannot obtain enough data for accurate assessment. Given that a highway concession is usually for a period of 25 years, setting the time point in the 8th operation year to the 10th operation year ($n \approx T/3$) is considered to be a proper choice. In this case, we collect operating data of ten years from Chang Jiu highway to carry out the mid-term risk assessment. Table 3 shows the operating revenue R^r_t and cost C^r_t from 2008 to 2018 and the data comes from the annual reports of the operator JIANGXI GANYUE EXPRESSWAY CO.,LTD.

Assume that both annual revenue and cost have linear growth pattern and obtain their first-order difference. ADF test in MATLAB for series ΔR_t and ΔR_t shows that they are stationary sequences.

According to Equation (7) and (8), we obtain the value of Kendall's tau of the two series $\tau_s = 0.4667$. Then, we can estimate the value of parameter θ for all the Copulas function: θ (Clayton)=1.75, θ (Frank) = (2.43, 5.06), θ (Ali-Mikhail-Haq) = 0.774, θ(Gumble Hougard) = 1.875.

Table 3 Operating Data of CJ Highway (2008-2018)

Year	2008	2009	2010	2011	2012	2013	2014
Operating Revenue ($R_t^r \times 10^4 CNY$)	69912.8	73479.3	85017.3	79702.3	75600.4	67003.1	73913.3
Operating Cost ($C_t^r \times 10^4 CNY$)	20921.7	21766.94	32652.58	28571.01	27399.6	27901.01	32694.4
Year	2015	2016	2017	2018			
Operating Revenue	79863.3	77814.2	81546.7	77890			
Operating Cost	43208.9	34934.6	32727.9	33051.8			

5.2 Choice of Copula Function

Once the parameter θ is determined, the generator φ_θ is obtained. The next step is to generate bivariate distribution $(\Delta R_t^s, \Delta C_t^s)$ through above instructions. The marginal distribution F_R and F_C of ΔR_t and ΔC_t are empirical distribution of them based on data from 2008 to 2018. When random number u and v are generated, the corresponding ΔR_t^s, ΔC_t^s can be worked out through empirical distribution F_R and F_C. However, since the empirical distribution functions are staircase functions with limited data points, linear interpolation is applied here to get simulated points as shown in Figure 4. For instance, if $u = 0.55$, the corresponding $\Delta R = 0.0759$ can be obtained through linear interpolation of the existing points (-0.2049,0.5) and (0.3567,0.6).

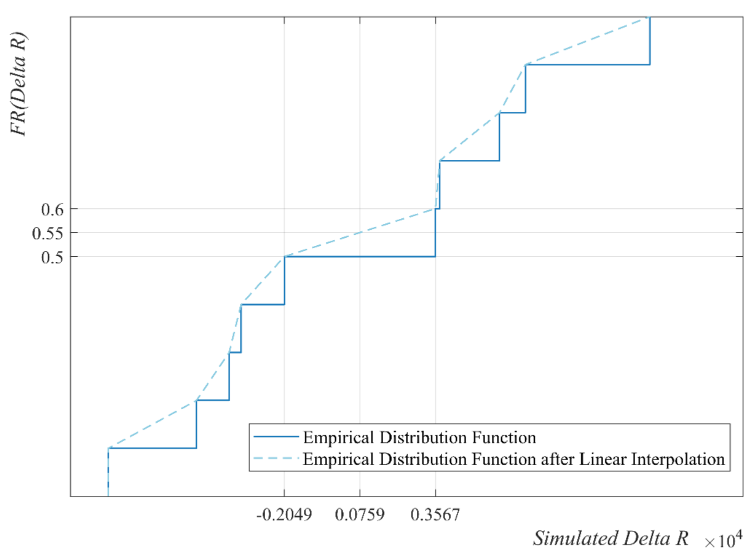

Figure 4 Obtaining ΔR_t^s through Linear Interpolation

Given the generated $(\Delta R_t^s, \Delta C_t^s)$, the simulated NPV of these ten years $NPV_{[1,10],i}^s$, average number M and degree of deviation D are worked out in turn. Figure 5 illustrates the simulated $NPV_{[1,10],i}^s$ scatter grams with different Copulas and without Copula and Table 4 specifies their M and D.

The NPV of practical data in this ten-year operation period, discounted at 5%, equals to 3.8898 billion CNY. According to Table 4 and Figure 5, the simulations based on Frank Copula and Gumble Hougard Copula show the best accuracy in average level of simulated numbers, because their M values are closer to $NPV_{[1,n]}^r$. Besides, the simulations based on Clayton Copula and Gumble Hougard Copula show the minimal degree of deviation with minimal D.

The simulation without Copula is presented as a control group, in which ΔR and ΔC are treated as independent variables. All the simulation results with Copula show an average NPV lower than real NPV, while that of control group is the only one larger than $NPV_{[1,n]}^r$. This is mainly because if the relevance of annual revenue and cost is taken into account, the distribution of NPV follows a fat-tailed pattern which is not the case for independent variables. Besides, the simulated NPV in control group has an obviously larger degree of statistical dispersion than NPV generated with Copulas.

Therefore, simulations with Copula proved to be a better way to get the distribution of NPV in accordance with actual conditions. According to the M and D in Table 4, Gumble Hougard Copula is chosen as the best way to describe the relevance of annual cost and revenue.

Table 4 Accuracy of Simulation with Different Copulas and without Copula

Copula	$M(\times 10^4 CNY)$	D
Clayton ($\theta = 1.75$)	3.7133e+05	0.0864
Frank ($\theta = 2.43$)	3.8619e+05	0.1017
Frank ($\theta = 5.06$)	3.8707e+05	0.1274
Ali-Mikhail-Haq ($\theta = 0.774$)	3.7623e+05	0.1226
Gumble Hougard ($\theta = 1.875$)	3.8441e+05	0.0901
N/A	3.9820e+05	0.1814

5.3 Calculation of NPVaR

Finally, the simulated NPV are generated. Figure 6 is the probability distribution and cumulated distribution of simulated NPV based on 1,000 trials (N=1,000). The NPVaRs at different confidence levels are marked on the axis.

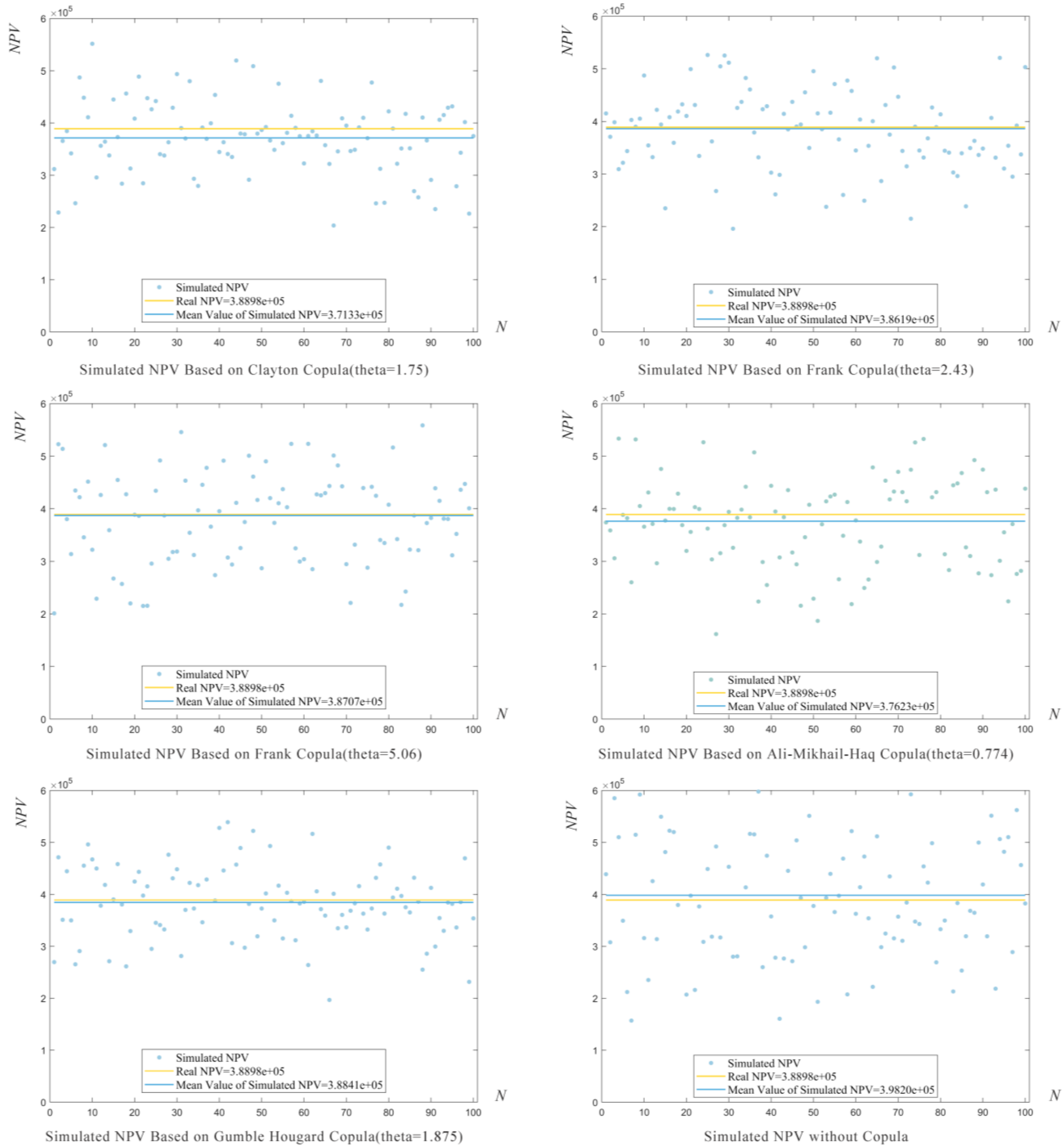

Figure 5 Scatter Grams of simulated NPV (100 Trials) with different Copulas and without Copulas

The purpose of this mid-term assessment is to provide a quantified reference for renegotiations or other dynamic adjustments in operation period, as a result of risk sharing. NPVaR measures the minimal possible value at a certain confidence level based on the practical operating data. For instance, 6.1647 billion CNY is a very pessimistic estimate of NPV_o at the confidence level of 0.95. Given an expected NPV_o of 7 billion CNY, $NPVaR_{0.95}$ implies that the risks in operation period may cause 0.8353 billion CNY loss in NPV. If the public sector and private sector share the risk equally, each of them should take a loss of 0.41765 billion CNY. In the same way, NPVaR can also measure the possible excess profit of this PPP project, as a quantified reference of MRC. This situation happens when the confidence level is assigned as less than 0.5, where $NPVaR_\alpha$ represents a very optimistic estimate of NPV_o. As shown in Table 4, given the expected NPV_o of 7 billion CNY, $NPVaR_{0.05}$ implies an excess profit of 0.8818 billion CNY, and both sectors can get 0.4409 billion CNY extra profit based on 1:1 sharing ratio.

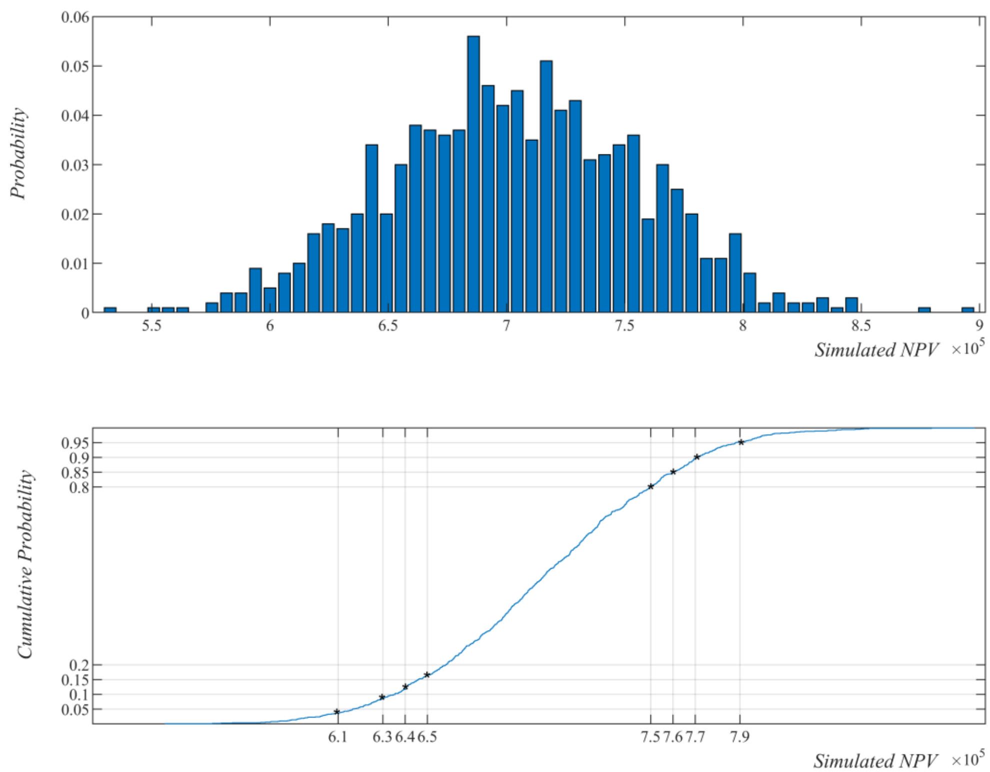

Figure 6 Simulation Results Based on Gumble Hougard Copula

6 Discussions

Numerous existing studies has focused on the revenue risk assessment model for ex-ante analysis, but the improved model for mid-term assessment has been scarcely addressed due to the lack of awareness of life-cycle management and efficient analytic tools. Therefore, new assessment models should be established to make use of the advantage of mid-term assessment: the project produces practical data to help appraisers make more realistic and reasonable assumptions in the assessment model rather than experience-based hypothesis. The Copula-NPVaR model proposed in this paper provides a method to present the interdependent variables in joint distribution and produces more accurate simulation results. Based on the numerical simulation results in case study, some suggestions and remarks for the implementation of better mid-term revenue risk assessment are summarized as follows:

(1) The relationship between interdependent variables cannot be neglected in the mid-term assessment model. According to the simulation results, if the annual income and annual cost are treated as independent variables as the control group, the simulated NPVs show in a more scattered way with more extreme values. Specifically, the pessimistic estimation will probably show NPV in a lower level and the optimistic estimation show larger NPV, which misleads the stakeholders into overestimating the revenue risk.

(2) The revenue floor and cap in MRG & MRC should not be immutable in the operation period. The NPV distribution simulated in mid-term assessment may considerably deviate from that based on ex-ante assumptions. If we cannot distinguish which sector is responsible for this deviation, it is unfair to still apply the previous revenue floor and cap. For instance, if the project gets more revenue mainly due to high quality of service provided by the private sector, a low cap may strip the revenue from it, and if the project gets less revenue mainly because of the poor management, then a high floor makes the public sector undertake unnecessary revenue risk. The mid-term assessment provides quantified reference for stakeholders to know about how much loss or abnormal profit they would have.

(3) A data analysis software may help with the implementation of the mid-term assessment. Some steps within this simulation model such as the random variable generation and the choice of Copula function involves complex calculations. Therefore, a data analysis software, comprising data encapsulation, is supposed to be developed to simplify the operation of appraisers.

(4) The Copula-NPVaR model can also be applied to mid-term revenue risk assessment or project evaluation in any other fields marked by long concession period and interdependent cash flow divisions. As long as enough operating data is available, appraisers can use this model to re-evaluate the revenue risk.

7 Conclusions

Mid-term revenue risk assessment in operation period holds great importance for highway PPP projects, enhancing the life-cycle risk management in concession period. Among various risk assessment methods, NPV-at-Risk analysis is the most intuitive and easy-to-operate method. However, how to employ the operating data in mid-term assessment model remains unclear. This paper proposed an improved model, applying

Copula functions to establish bivariate distribution of revenue and cost based on cash flow data of the project appraised. The main contribution is to find a way to make more reasonable stochastic assumptions on interdependent variables in the simulation of NPV-at-Risk. The simulation results in case study shows that the Copula-NPVaR model can reduce the statistic dispersion of simulated NPVs and provide quantified reference more consistent with the real situation. The present study has been one of the first attempts to indicate the relationship between variables in the NPV simulation model via mathematical tools and these findings can promote the project managers to carry out the mid-term revenue assessment in a more scientific way.

Despite the academic and practical contributions, the limitation of this study is that it only applies bivariate Copula and divides the cash flow simply into inflow and outflow without breaking down the income and expenditure. In reality, some cost divisions and income divisions may be more relative with each other while others are independent. Therefore, future work can focus on applying multivariate Copula or other data analysis tool to embody the multivariate relationship in this simulation model if more detailed data is available.

Author Contribution Statement

The authors confirm contribution to the paper as follows: 1) Study conception and design: Zhipeng Zhang, Hao Hu, Yujie Huang; 2) Data collection, analysis and interpretation of results: Yujie Huang, Lei Dai, Jie Xue; 3) Draft manuscript preparation: Yujie Huang, Zhipeng Zhang, Hao Hu. All authors reviewed the results and approved the final version of the manuscript.

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